**NATIONAL TEACHERS COLLEGE**
*629 J. Nepomuceno St, Quiapo Manila, 1001 Metro Manila*

## *Activity Lab 7*

# *"Robust Inventory Form with Real-Time Validation and Validation Trace Log"*

In Partial Fulfillment of the Requirements

in Python Programming for the Degree of

Bachelor of Science in Information Technology

**Submitted by:**

**Romiel Macawiwili** - GUI design and window layout

**John Michael Tolentino** - Real-time validation using StringVar() and trace()

**Sean Lawrence Tapangan** - Error handling using try-except

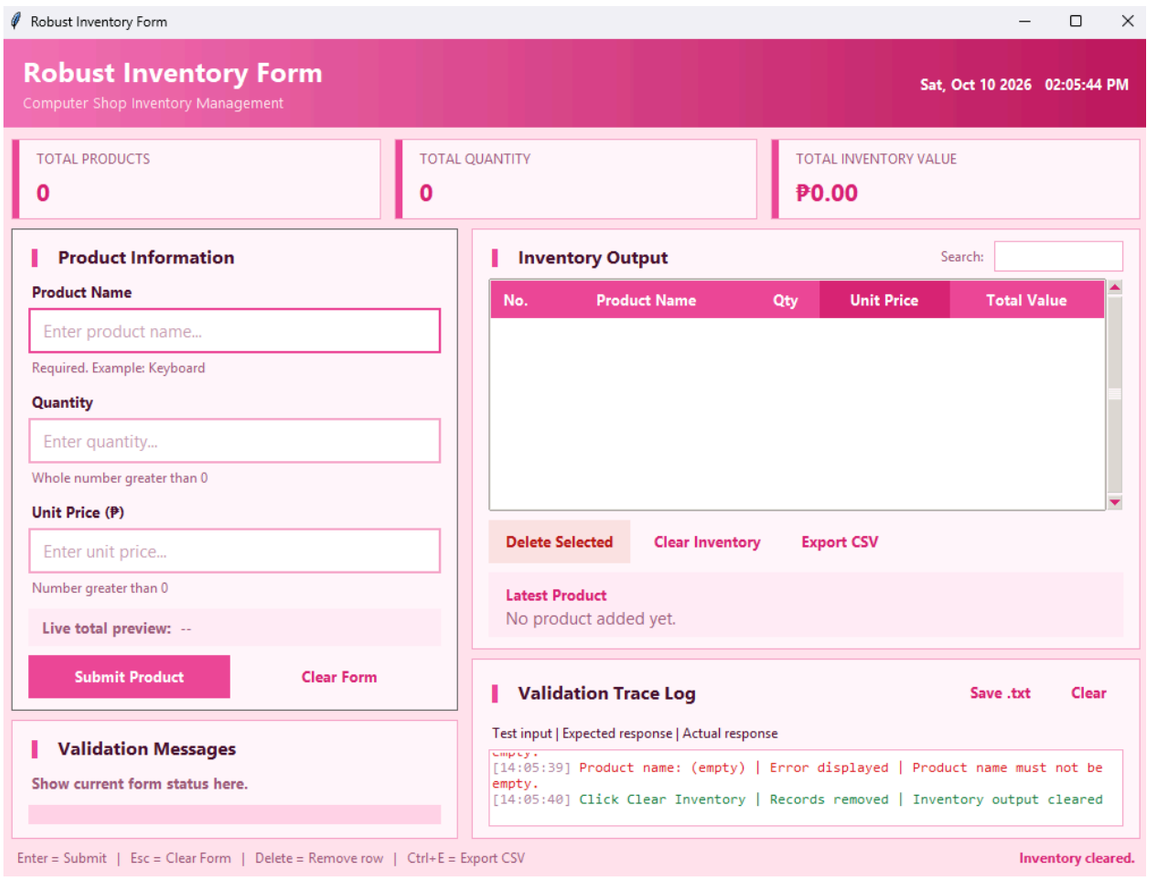
**Rhaily Bantug** - Testing and Validation Trace Log

**Diana Shane Aguilon** - Documentation and demonstration

Submitted to:
**Mr. Jasper Kutz Ebedo**

**Date of Submission:**
October 10, 2026

## Task 1 GUI

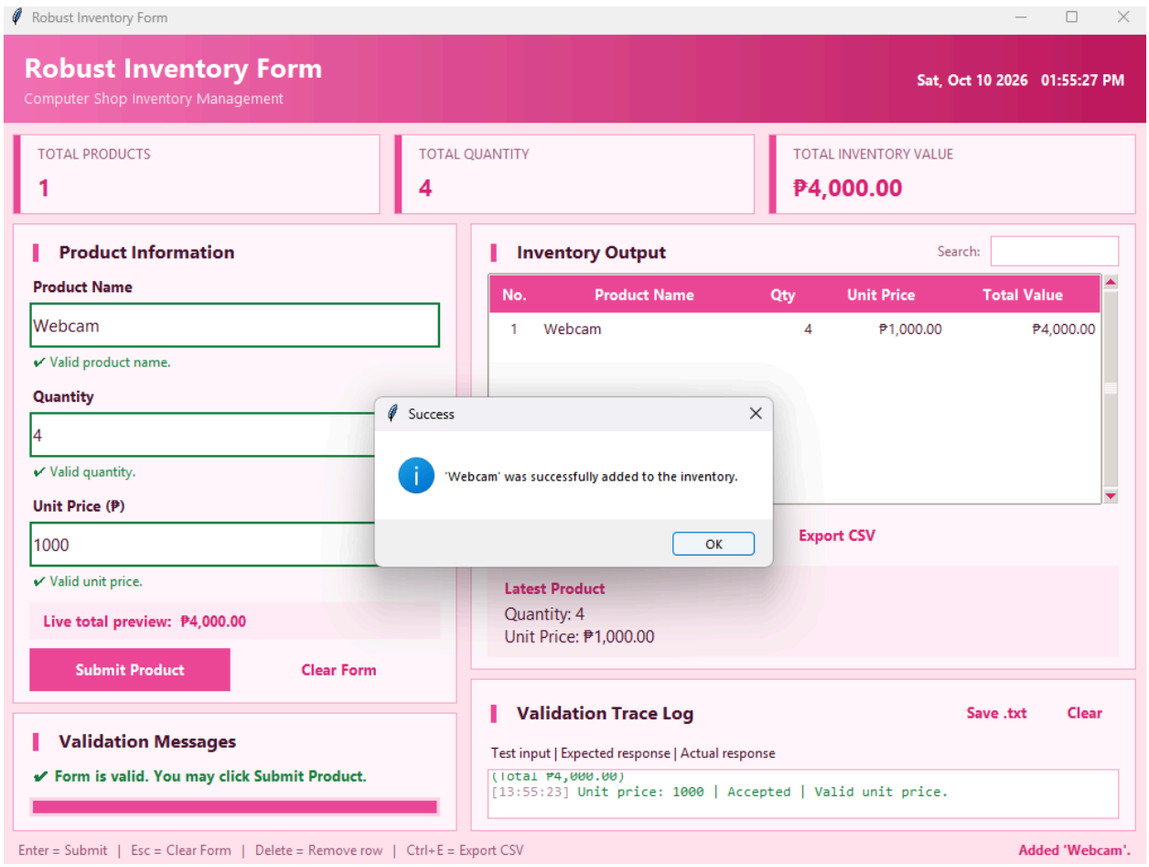



## Validation Trace Log

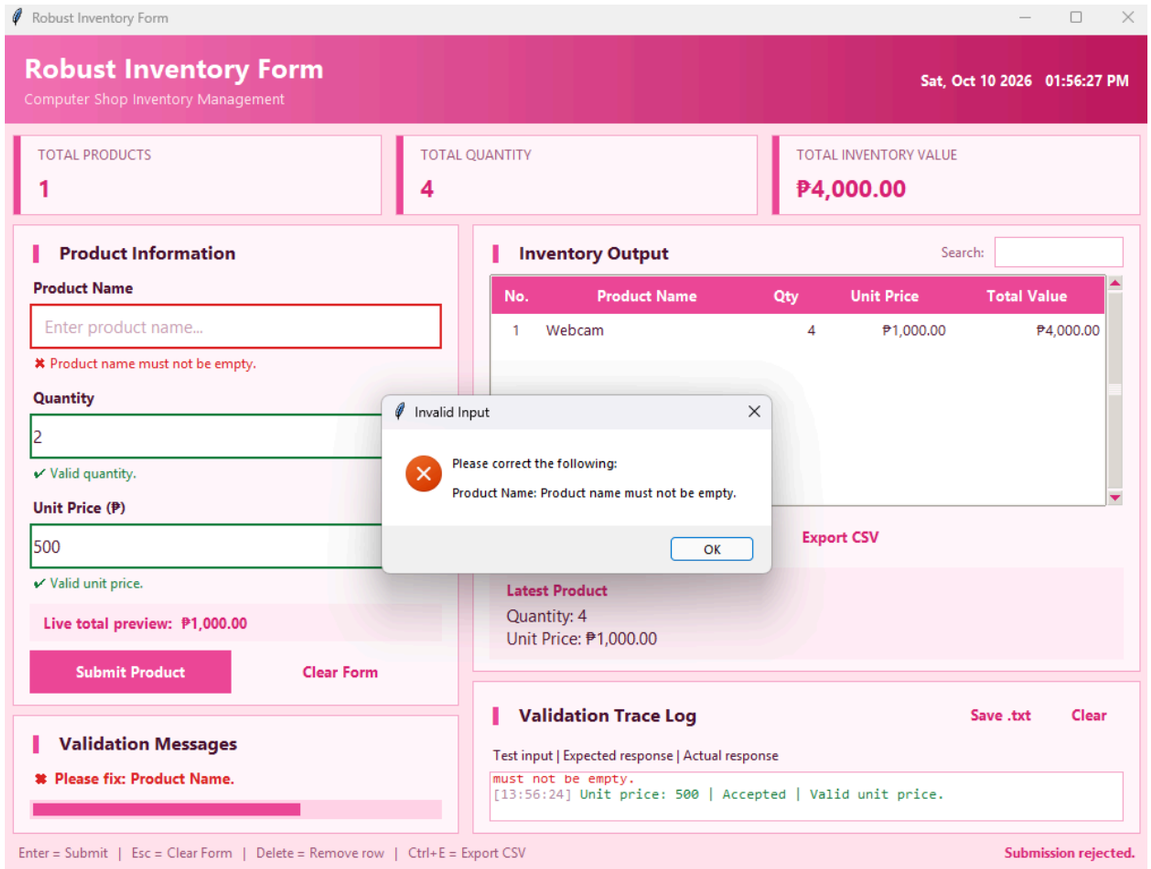
| No. | Test Input / Action | Expected Result | Actual Result |
|---|---|---|---|
| 1 | Valid product name | Accepted | Product is successfully added. Total value is PHP 4,000. |
| 2 | Empty product name | Error displayed | Validation appears for the product name. |
| 3 | Quantity: abc | Rejected | Validation appears for the quantity. |
| 4 | Quantity: 0 | Rejected | Validation appears for the quantity because it must be whole numbers. |
| 5 | Quantity: -3 | Rejected | Validation appears for the quantity because it must be positive. |
| 6 | Quantity: 5.5 | Rejected | Quantity is rejected because it must be a whole number. |
| 7 | Unit price: hello | Rejected | Validation error appears for the unit price. |
| 8 | Unit price: -100 | Rejected | Negative price is rejected. |
| 9 | Submit all valid inputs | Product added | All valid products successfully added. |

| | | | |
|---|---|---|---|
| 10 | Click Clear Form | Fields reset | Input fields are cleared. Existing inventory records remain. |

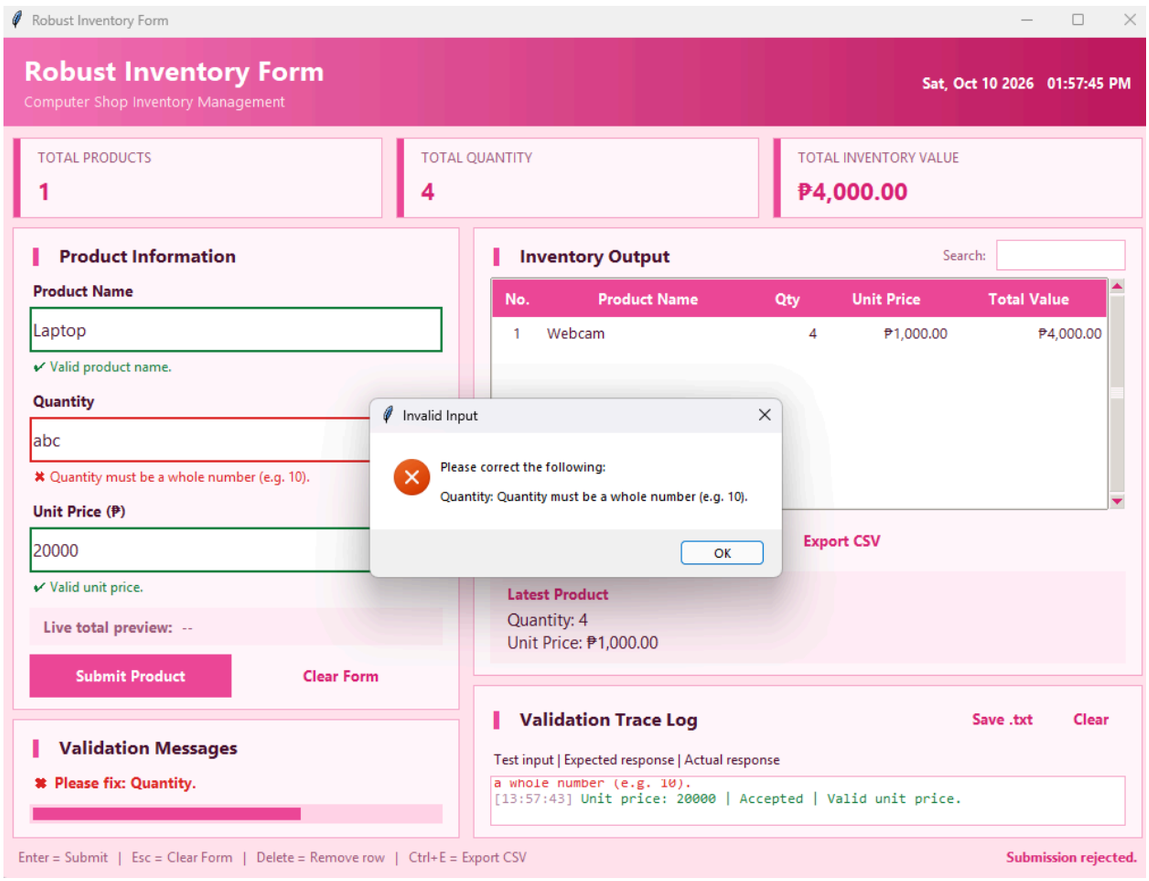
**Test 1**



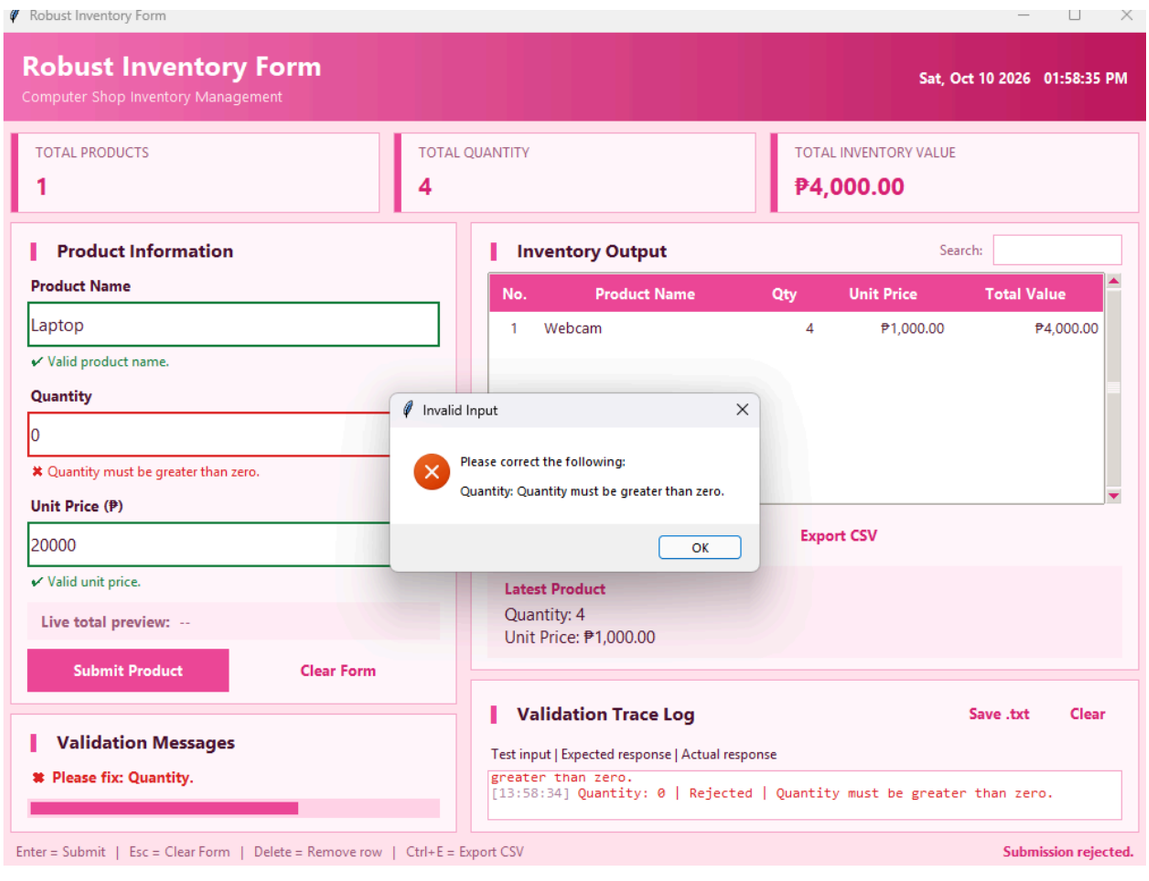

**Test 2**



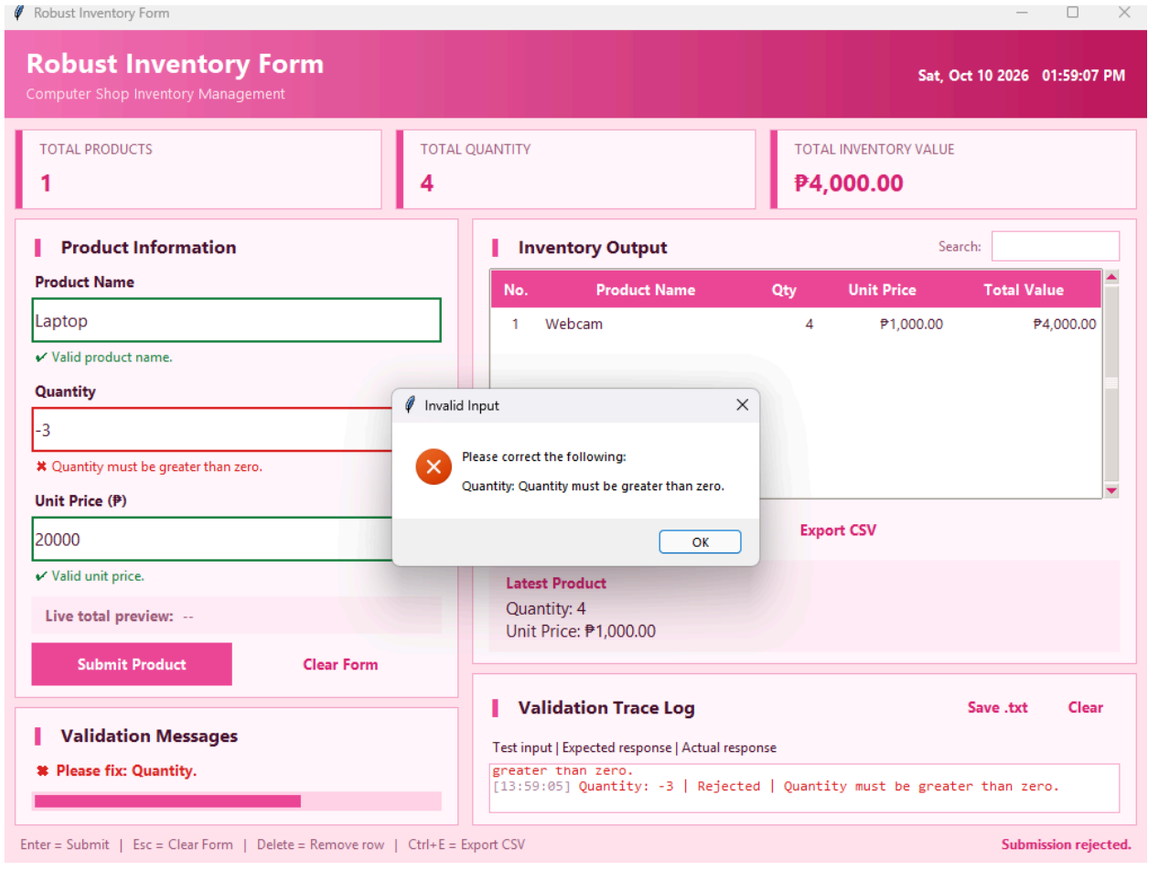
## Test 3


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 01:57:45 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
abc
✖ Quantity must be a whole number (e.g. 10).
Unit Price (₱)
20000
✔ Valid unit price.
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Quantity.
Inventory Output
Search:
No. Product Name Qty Unit Price Total Value
1 Webcam 4 ₱1,000.00 ₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Invalid Input
Please correct the following:
Quantity: Quantity must be a whole number (e.g. 10).
OK
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
a whole number (e.g. 10).
[13:57:43] Unit price: 20000 | Accepted | Valid unit price.
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

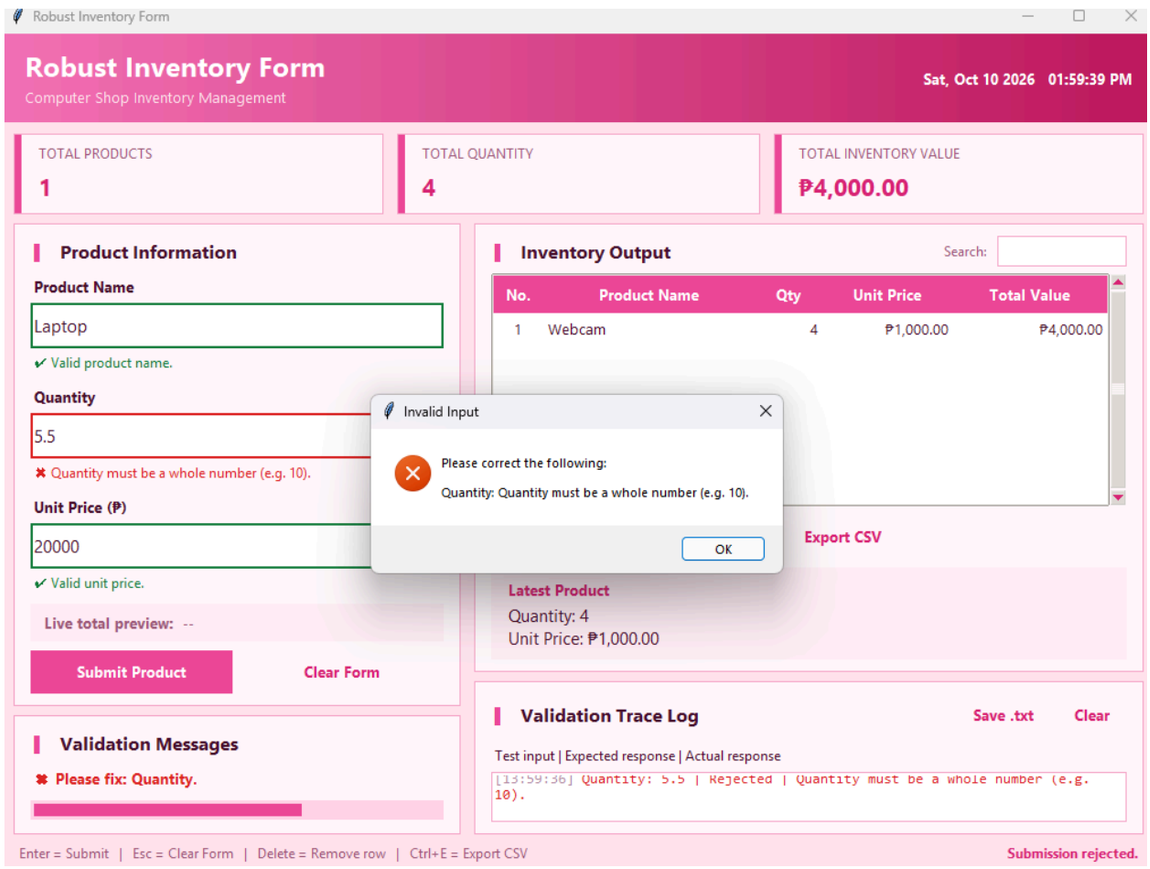
## Test 4


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 01:58:35 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
0
✖ Quantity must be greater than zero.
Unit Price (₱)
20000
✔ Valid unit price.
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Quantity.
Inventory Output
Search:
No. Product Name Qty Unit Price Total Value
1 Webcam 4 ₱1,000.00 ₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Invalid Input
Please correct the following:
Quantity: Quantity must be greater than zero.
OK
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
greater than zero.
[13:58:34] Quantity: 0 | Rejected | Quantity must be greater than zero.
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

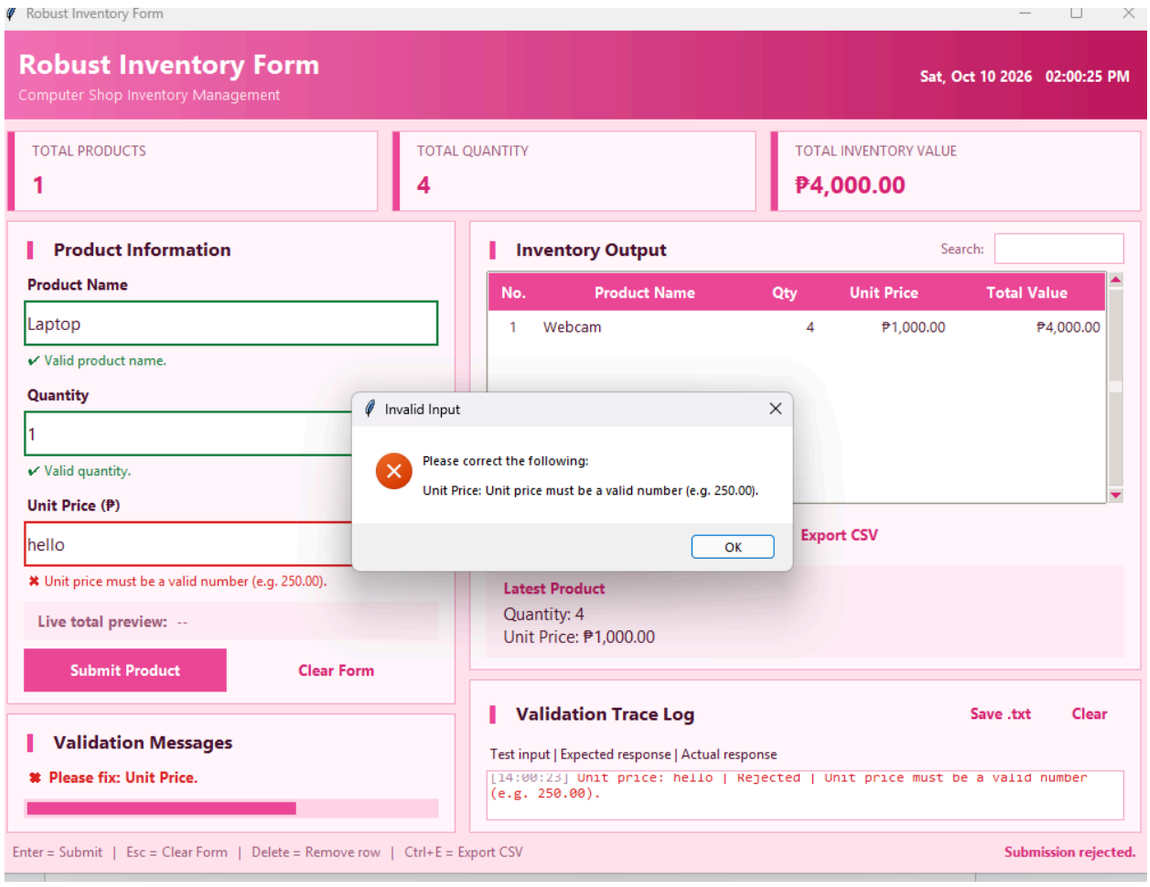
## Test 5


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 01:59:07 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
-3
✖ Quantity must be greater than zero.
Unit Price (₱)
20000
✔ Valid unit price.
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Quantity.
Inventory Output
Search:
No. Product Name Qty Unit Price Total Value
1 Webcam 4 ₱1,000.00 ₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
greater than zero.
[13:59:05] Quantity: -3 | Rejected | Quantity must be greater than zero.
Invalid Input
Please correct the following:
Quantity: Quantity must be greater than zero.
OK
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

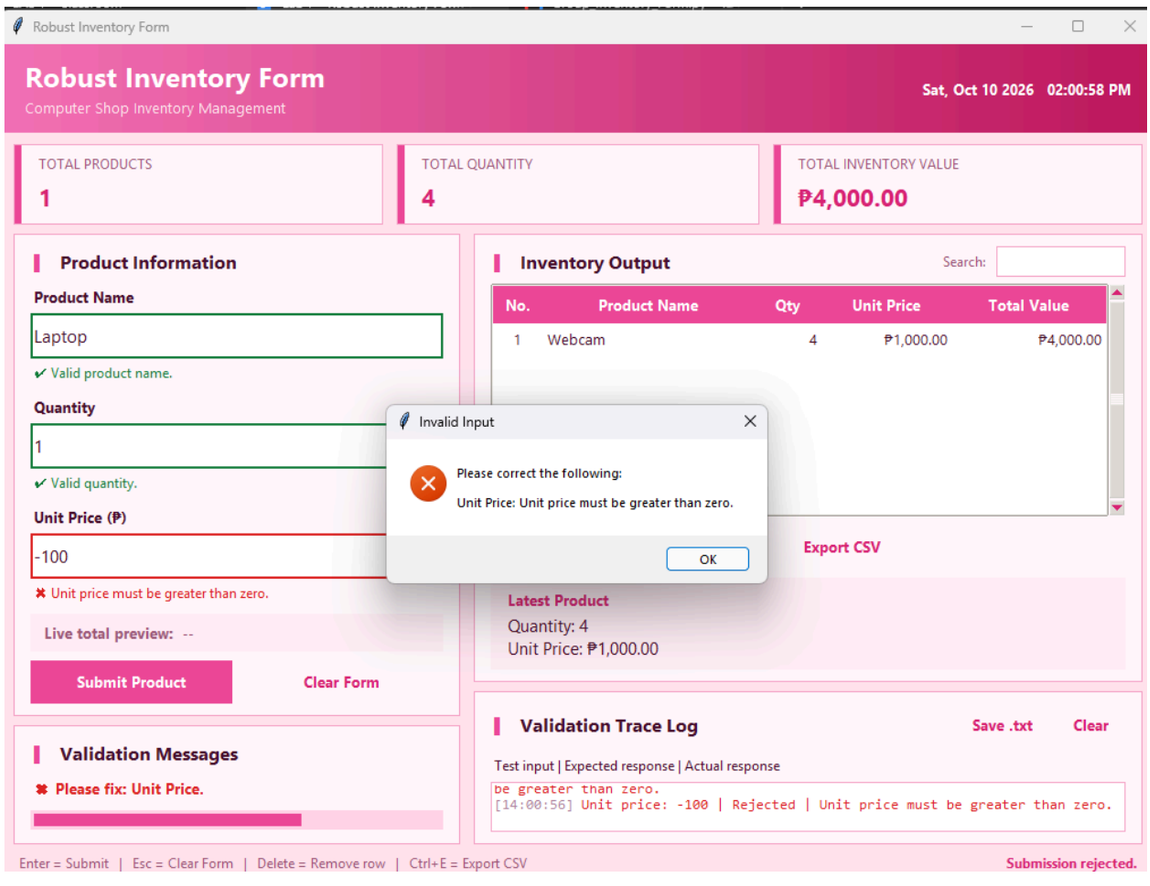
## Test 6


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 01:59:39 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
5.5
✖ Quantity must be a whole number (e.g. 10).
Unit Price (₱)
20000
✔ Valid unit price.
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Quantity.
Inventory Output
Search:
No. Product Name Qty Unit Price Total Value
1 Webcam 4 ₱1,000.00 ₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
[13:59:36] Quantity: 5.5 | Rejected | Quantity must be a whole number (e.g. 10).
Invalid Input
Please correct the following:
Quantity: Quantity must be a whole number (e.g. 10).
OK
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

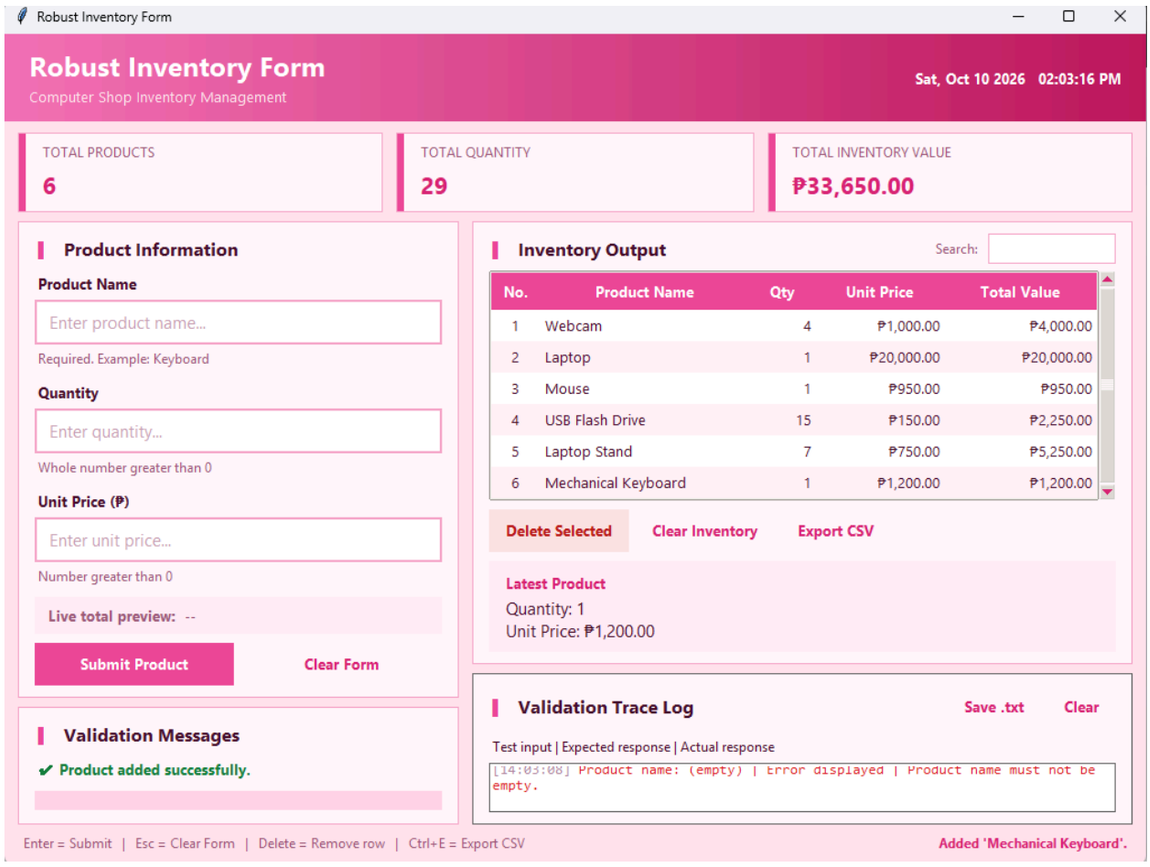
## Test 7


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 02:00:25 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
1
✔ Valid quantity.
Unit Price (₱)
hello
✖ Unit price must be a valid number (e.g. 250.00).
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Unit Price.
Inventory Output
Search:
No.
Product Name
Qty
Unit Price
Total Value
1
Webcam
4
₱1,000.00
₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Invalid Input
Please correct the following:
Unit Price: Unit price must be a valid number (e.g. 250.00).
OK
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
[14:00:23] Unit price: hello | Rejected | Unit price must be a valid number (e.g. 250.00).
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

## Test 8


Robust Inventory Form
Robust Inventory Form
Computer Shop Inventory Management
Sat, Oct 10 2026 02:00:58 PM
TOTAL PRODUCTS
1
TOTAL QUANTITY
4
TOTAL INVENTORY VALUE
₱4,000.00
Product Information
Product Name
Laptop
✔ Valid product name.
Quantity
1
✔ Valid quantity.
Unit Price (₱)
-100
✖ Unit price must be greater than zero.
Live total preview: --
Submit Product
Clear Form
Validation Messages
✖ Please fix: Unit Price.
Inventory Output
Search:
No.
Product Name
Qty
Unit Price
Total Value
1
Webcam
4
₱1,000.00
₱4,000.00
Export CSV
Latest Product
Quantity: 4
Unit Price: ₱1,000.00
Invalid Input
Please correct the following:
Unit Price: Unit price must be greater than zero.
OK
Validation Trace Log
Save .txt
Clear
Test input | Expected response | Actual response
be greater than zero.
[14:00:56] Unit price: -100 | Rejected | Unit price must be greater than zero.
Enter = Submit | Esc = Clear Form | Delete = Remove row | Ctrl+E = Export CSV
Submission rejected.

## Test 9




## Task 10

